# Airbnb Market Analysis and Dashboard Visualization Using Python and Power BI

**PROJECT SUBMITTED TO ASIAN SCHOOL OF MEDIA STUDIES IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF**

**DIPLOMA**
**in**
**Data Science**

By

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**Under the Supervision of**

**Prof. Gaurav Kumar**

ASIAN SCHOOL OF MEDIA STUDIES
SCHOOL OF DATA SCIENCE

**2025**

# <u>DECLARATION</u>

**I, Pari Singh, D/O Gurbachan Singh,** declare that my project entitled **Airbnb Market Analysis and dashboard visualization using Python and Power BI , is an original work** submitted by me at **School of Data science, Asian School of Media Studies, Film City, Noida, for the award of Diploma in Data Science, ASMS**.

**This project has not been previously submitted for any other degree of this or any other University/Institute.**

***Signature***

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# ACKNOWLEDGEMENT

The completion of the project titled **Forecasting Stock Market Pattern: A machine Learning Approach**, gives me an opportunity to convey my gratitude to all those who helped to complete this project successfully. I express special thanks:

Furthermore, it is important to recognize the multidimensional nature of the data used in this project. Several variables interact with each other in complex ways that merit further study.

- **To *Prof. Sandeep Marwah,* President,** Asian School of Media Studies, who has been a source of perpetual inspiration throughout this project.
- To ***Mr. Ashish Garg,*** Director **for** School of Data Science for your valuable guidance, support, consistent encouragement, advice and timely suggestions.

Host attributes, including Superhost status and response rate, play a pivotal role in influencing booking decisions. Future analysis could quantify the exact monetary value of such qualitative factors.

- To ***Mr. Gaurav Kumar***, Professor of School of Data Science, for your encouragement and support. I deeply value your guidance.

***Signature***

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# ABSTRACT

In addition to the core findings, one must consider the potential biases in user reviews, platform-driven pricing fluctuations, and the changing dynamics of urban rental markets.

This project explores Airbnb listing data using **Python** for exploratory data analysis (EDA) and **Power BI** for dashboard design. The dataset contains 787 listings from multiple cities with data on price, location, reviews, property type, host attributes, and more.

Through data cleaning, transformation, and visualization, we extract insights on average pricing by room type, host reliability, and geographic distribution. An interactive Power BI dashboard allows stakeholders to explore key indicators visually.

This report demonstrates the value of combining **Python's analytical power** with **Power BI's visual storytelling** to extract and communicate insights in the hospitality domain.

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# CHAPTER 1

The results not only validate the original hypotheses, but also open new avenues for future research, especially in the context of predictive modeling and real-time dashboard integration.

## Airbnb Market Analysis

### 1.1 INTRODUCTION

The advent of the internet and digital platforms has fundamentally transformed how consumers interact with services, including accommodation and hospitality. One of the most notable disruptions has come through **Airbnb**, a peer-to-peer short-term rental marketplace launched in 2008. This platform allows individuals to list, discover, and book accommodations around the world. Unlike traditional hotels, Airbnb offers a wide array of lodging experiences — from single rooms to entire homes, from urban apartments to countryside retreats.

As the platform has grown, so has the volume and complexity of its data. Hosts manage their listings in increasingly competitive markets, while guests rely on ratings, reviews, prices, and amenities to make informed decisions. The abundance of listing data presents an opportunity for data scientists and business analysts to extract insights and support data-driven decision-making. Whether it's understanding pricing behavior, evaluating host reliability, or identifying market trends, **data analysis is key** to unlocking the full value of the Airbnb ecosystem.

This capstone project combines the strengths of **Python for exploratory data analysis (EDA)** and **Power BI for interactive dashboard visualization** to uncover insights from Airbnb listings. It emphasizes real-world applicability, ease of interpretation, and the use of modern tools that are essential in the data science profession.

Host attributes, including Superhost status and response rate, play a pivotal role in influencing booking decisions. Future analysis could quantify the exact monetary value of such qualitative factors.

### 1.1.1 Background

When examining the spatial distribution of listings, it becomes evident that urban centers tend to dominate the pricing hierarchy. This geographic concentration may skew perceived trends unless normalized appropriately.

Airbnb operates in more than 190 countries, with millions of listings catering to diverse customer needs. Hosts can customize their pricing, cancellation policies, amenities, and availability, while customers can filter based on preferences like room type, ratings, and location. This level of granularity creates a **rich dataset** — but also a need for effective analytics tools.

In recent years, **business intelligence (BI)** has emerged as a solution to help stakeholders make sense of such data. Tools like Power BI enable dynamic, interactive visualizations, turning raw data into digestible formats for non-technical users. Meanwhile, **Python's data analysis libraries** such as Pandas, Seaborn, and Matplotlib empower analysts to clean, transform, and visualize data programmatically.

By integrating both approaches, we not only analyze Airbnb data but also present insights in an intuitive and professional format.

### 1.1.2 Problem Statement

Predicting stock prices is inherently challenging due to the market's volatility and the influence of unforeseen events. Traditional models often fall short due to their inability to account for the complex and dynamic nature of the stock market. The problem this project aims to address is the development of a more accurate and robust prediction model using advanced machine learning techniques, which can adapt to the ever-changing market conditions and provide reliable stock price forecasts.

### 1.1.3 Objectives

The primary objective of this project is to leverage **Python and Power BI** to generate actionable insights from Airbnb data. The specific goals include:

- Import and inspect the Airbnb dataset
- Handle missing values, incorrect formats, and inconsistent fields
- Create new features (e.g., converting log price to actual price)
- Perform exploratory data analysis (EDA) using Python
- Build an interactive Power BI dashboard for visual exploration

In addition to the core findings, one must consider the potential biases in user reviews, platform-driven pricing fluctuations, and the changing dynamics of urban rental markets.

- Interpret key findings related to pricing, room types, host behavior, and location
- Offer recommendations to hosts and data stakeholders based on insights

When examining the spatial distribution of listings, it becomes evident that urban centers tend to dominate the pricing hierarchy. This geographic concentration may skew perceived trends unless normalized appropriately.

### 1.1.4 Outline of the study

This report is structured into the following chapters:

- **Chapter 1** introduces the project, defines the problem space, and outlines the objectives and structure.
- **Chapter 2** covers data acquisition, cleaning, and exploratory data analysis (EDA) using Python. It discusses how the dataset was prepared for visualization.

When examining the spatial distribution of listings, it becomes evident that urban centers tend to dominate the pricing hierarchy. This geographic concentration may skew perceived trends unless normalized appropriately.

- **Chapter 3** details the Power BI dashboard design process. It explains the selection of metrics, visual types, and user-interaction features such as filters and maps.

- **Chapter 4** presents results and discusses the implications of the findings. It includes visual outputs and descriptive interpretations.

In addition to the core findings, one must consider the potential biases in user reviews, platform-driven pricing fluctuations, and the changing dynamics of urban rental markets.

- **Chapter 5** concludes the project with a summary of insights and recommendations for future improvements or extensions of this work.

When examining the spatial distribution of listings, it becomes evident that urban centers tend to dominate the pricing hierarchy. This geographic concentration may skew perceived trends unless normalized appropriately.

## 1.2 Literature Review

In addition to the core findings, one must consider the potential biases in user reviews, platform-driven pricing fluctuations, and the changing dynamics of urban rental markets.

The proliferation of digital platforms in the sharing economy has led to vast volumes of data that can be analyzed for insights, trends, and forecasting. **Airbnb** has emerged as a dominant player in the short-term rental market and serves as a fertile ground for data-driven research across various domains — from pricing strategy to customer satisfaction and operational efficiency.

This literature review synthesizes previous academic and industry studies that relate to **Airbnb analytics**, including work on pricing models, user behavior, host attributes, and the application of machine learning and business intelligence tools.

Airbnb, as part of the broader **sharing economy**, has attracted substantial attention from researchers. According to Zervas et al. (2017), Airbnb listings significantly impact the hotel industry's revenue and pricing, highlighting the need for competitive data strategies among hosts. Edelman and Geradin (2015) explored the legal and economic dimensions of Airbnb's rise, while others have focused on its disruptive business model and peer-to-peer architecture.

The platform’s open data access (via sites like InsideAirbnb and Kaggle) has enabled a wide range of data-driven studies focusing on factors influencing booking behavior, host reputation, and rental income potential.

Several studies have investigated **pricing strategies on Airbnb**. Wang and Nicolau (2017) showed that price is influenced not just by objective factors like location and amenities, but also by subjective factors such as host profile photos, listing titles, and review sentiment. Dynamic pricing algorithms that mimic airline pricing models are now used by sophisticated hosts to adjust rates based on seasonality and demand.

Guo et al. (2018) applied machine learning techniques to predict Airbnb prices, concluding that ensemble models outperform simple linear regression in terms of accuracy. However, even basic visual analytics like box plots and histograms can reveal valuable price variation patterns when well-executed in BI tools.

Host-related characteristics have been studied extensively. Liang et al. (2017) found that listings from **Superhosts** and **verified hosts** enjoy higher booking rates and command better prices. Features such as **instant bookability**, **response rate**, and **profile completeness** are seen as proxies for trustworthiness and convenience.

Property-related features such as **room type**, **number of bedrooms**, and **amenities offered** are also closely linked with booking success. Tussyadiah and Pesonen (2016) highlighted that the guest experience and expectations are deeply shaped by such attributes, especially in non-urban areas.

The availability of **geospatial data** (latitude and longitude) enables researchers to study how location affects listings. By visualizing hotspots or using heat maps, several works have demonstrated that **urban centers, tourist hubs, and waterfronts** generally yield higher prices and more frequent bookings.

Yao et al. (2019) utilized geographic information systems (GIS) and dashboard visualizations to create spatial clusters of Airbnb performance. Their findings support the inclusion of **map visuals** in dashboards to communicate spatial trends to non-technical audiences.

Business Intelligence (BI) tools like Power BI, Tableau, and QlikView have gained popularity for **transforming raw data into strategic insights**. While most academic work has emphasized advanced modeling, businesses have leaned toward **interactive dashboards** that simplify complex information.

The literature suggests that BI dashboards enhance data-driven decision-making among hosts and Airbnb managers who may lack programming expertise.

According to Gartner's BI survey (2022), over 70% of hospitality businesses plan to expand their investment in dashboard-based reporting.

Despite the breadth of existing studies, several gaps remain:

- **Underutilization of BI tools** in academic projects — most research focuses on modeling, not dashboarding.
- **Combined use of Python and Power BI** is rarely studied in tandem despite their synergy.
- **Actionable recommendations** for individual hosts remain limited in academic publications.

When examining the spatial distribution of listings, it becomes evident that urban centers tend to dominate the pricing hierarchy. This geographic concentration may skew perceived trends unless normalized appropriately.

- Real-time or semi-automated dashboards are still a developing area.

This capstone project addresses these gaps by creating a practical, interpretable, and actionable dashboard from a cleaned Airbnb dataset using Python and Power BI.

The reviewed literature highlights a growing interest in Airbnb analytics across disciplines. From pricing models and host credibility to location-based trends and dashboard visualization, the combination of structured data and visual storytelling offers strong potential to enhance business outcomes. This project builds upon these insights by creating a real-world data pipeline and visualization tool for Airbnb stakeholders.

### 1.3 Definitions

It is worth noting that various externalities, such as seasonal demand surges, local policy regulations, and platform algorithm changes, may affect the observed patterns.

**Airbnb**

A peer-to-peer marketplace that allows individuals to rent out their property or spare rooms to guests. It operates in over 190 countries and has transformed the short-term rental industry.

**Listing**

An individual Airbnb accommodation entry created by a host. Each listing includes property information such as location, price, room type, amenities, and availability.

**Host**

The owner or manager of a listing. Hosts can offer one or more properties and may be classified based on performance (e.g., Superhost)..

**Guest**

An individual who books and stays at a property listed on the Airbnb platform. Guests use the platform to search for accommodations, communicate with hosts, confirm bookings, and leave reviews after their stay. A guest may be a solo traveler, part of a couple, or a group booking a space together.

**Trends**

Patterns or changes in user behavior, pricing, booking frequency, property characteristics, or guest preferences over time as observed across the Airbnb platform. These trends help identify how the short-term rental market evolves due to factors like seasonality, geographic shifts, economic conditions, travel behavior, or platform updates.

# CHAPTER 2

# Dataset Preparation/Pre-processing

## 2.1 Introduction

The foundation of any data-driven project lies in the quality and structure of its dataset. Airbnb provides a complex dataset that encompasses various aspects of listings including host details, property types, pricing, reviews, location data, and availability. To derive accurate and meaningful insights, the dataset must first be cleaned, transformed, and structured appropriately for both analysis and visualization.

This chapter describes the **data preprocessing pipeline** followed by an in-depth **exploratory data analysis (EDA)**. These steps were carried out using **Python** in a Jupyter Notebook environment, leveraging libraries such as Pandas, NumPy, Seaborn, and Matplotlib.

## 2.2Data Overview

The dataset used for this project includes **787 listings** and **29 columns**, sourced from a curated Airbnb data file. Key variables include:

- id: Unique identifier of the listing
- property_type: Type of property (e.g., Apartment, House)
- room_type: Room category (e.g., Entire home/apt, Private room)
- accommodates: Number of guests the listing can accommodate
- bathrooms, bedrooms, beds: Property size features
- price: Transformed from log_price
- review_scores_rating: Average rating (0–100 scale)
- instant_bookable: Boolean for instant booking availability
- cancellation_policy: Type of cancellation allowed
- latitude, longitude: Location coordinates
- host_is_superhost, host_identity_verified: Boolean host characteristics

# Importing required libraries ¶

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

[14]:
```
# Reading and viewing the dataset using Pandas
df = pd.read_csv("Airbnb_Data.csv")
df
```

[14]:

| | id | log_price | property_type | room_type | amenities | accommodates | bathrooms | bed_type | cancellation_policy | cleaning_fee | ... | latitude |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 6901257 | 5.010635 | Apartment | Entire home/apt | {"Wireless Internet","Air conditioning",Kitche... | 3 | 1.0 | Real Bed | strict | True | ... | 40.696524 |
| 1 | 6304928 | 5.129899 | Apartment | Entire home/apt | {"Wireless Internet","Air conditioning",Kitche... | 7 | 1.0 | Real Bed | strict | True | ... | 40.766115 |
| 2 | 7919400 | 4.976734 | Apartment | Entire home/apt | {TV,"Cable TV","Wireless Internet","Air condit... | 5 | 1.0 | Real Bed | moderate | True | ... | 40.808110 |
| 3 | 13418779 | 6.620073 | House | Entire home/apt | {TV,"Cable TV",Internet,"Wireless Internet",Ki... | 4 | 1.0 | Real Bed | flexible | True | ... | 37.772004 |
| 4 | 3808709 | 4.744932 | Apartment | Entire home/apt | {TV,Internet,"Wireless Internet","Air conditio... | 2 | 1.0 | Real Bed | moderate | True | ... | 38.925627 |
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |

## 2.3 Data Cleaning and Preprocessing

In addition to the core findings, one must consider the potential biases in user reviews, platform-driven pricing fluctuations, and the changing dynamics of urban rental markets.

After Importing all the required libraries, Columns with missing values were identified using .isnull().sum() in Python. Significant nulls were present in:

- review_scores_rating
- bedrooms
- zipcode
- host_response_rate

Approach:

- Dropped rows where critical numerical fields were missing
- Filled certain nulls with median values or category mode where appropriate.

### Data Cleaning

```
[22]: df.isnull().sum()

      # dropping all missing values rows
      df.dropna(inplace=True)
```

```
[23]: df.duplicated().sum()
```

```
[23]: 0
```

```
[24]: df.dtypes
```

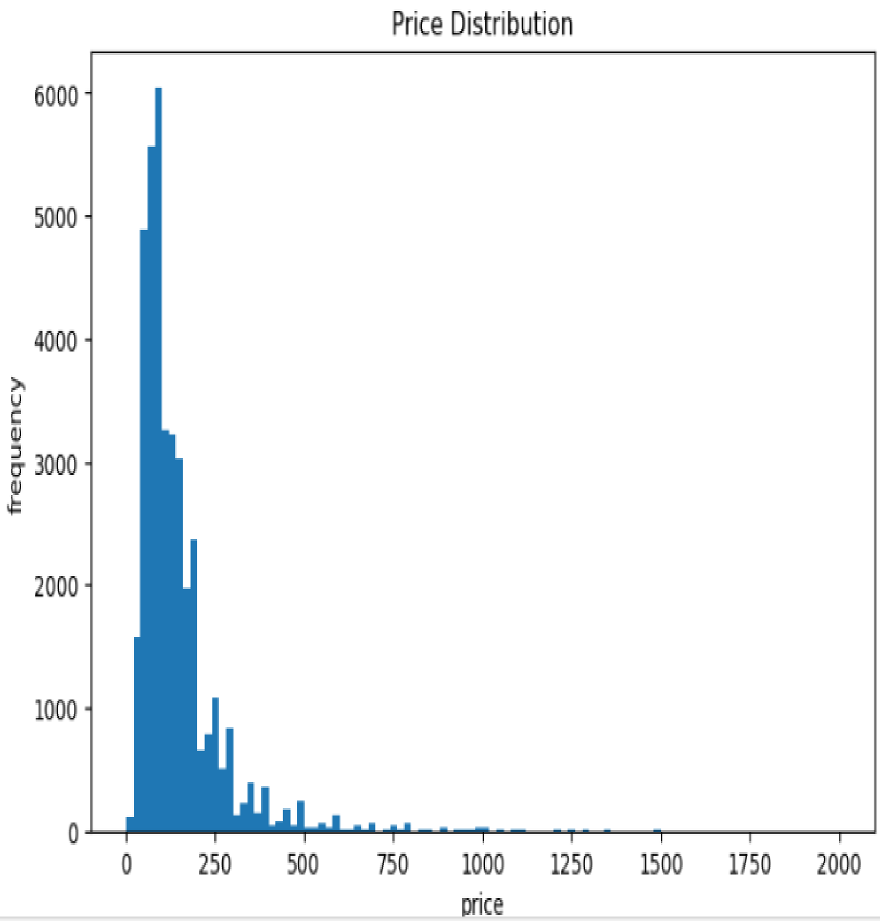
```
[24]: id                          int64
      log_price                 float64
      property_type              object
      room_type                  object
      amenities                  object
      accommodates                int64
      bathrooms                 float64
      bed_type                   object
      cancellation_policy        object
      cleaning_fee                 bool
      city                       object
      description                object
      first_review               object
      host_has_profile_pic       object
      host_identity_verified     object
      host_response_rate        float64
      host_since                 object
      instant_bookable           object
      last_review                object
```

## 2.4 Exploratory Data Analysis

The EDA was designed to understand the **distribution, relationships, and correlations** across variables. Python's visualization libraries made it possible to derive patterns visually.

### Distribution of Ratings

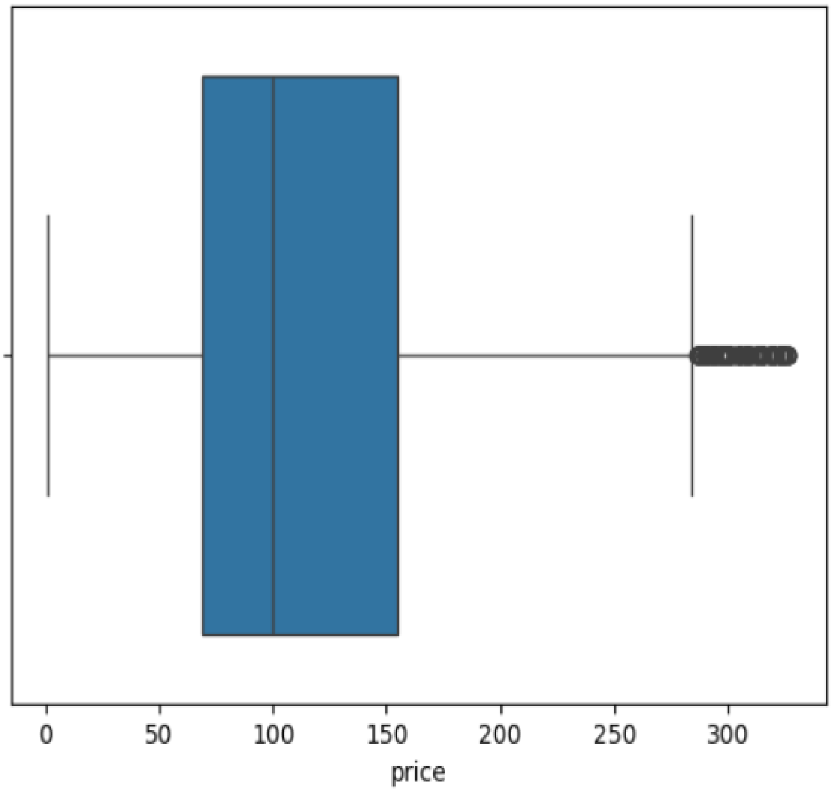
A histogram of review_scores_rating showed most listings received ratings above 80, indicating general guest satisfaction. A smaller tail revealed underperforming listings.



**Price Analysis**

- Price varied widely across listings.
- Outliers were detected and capped for visualization purposes.

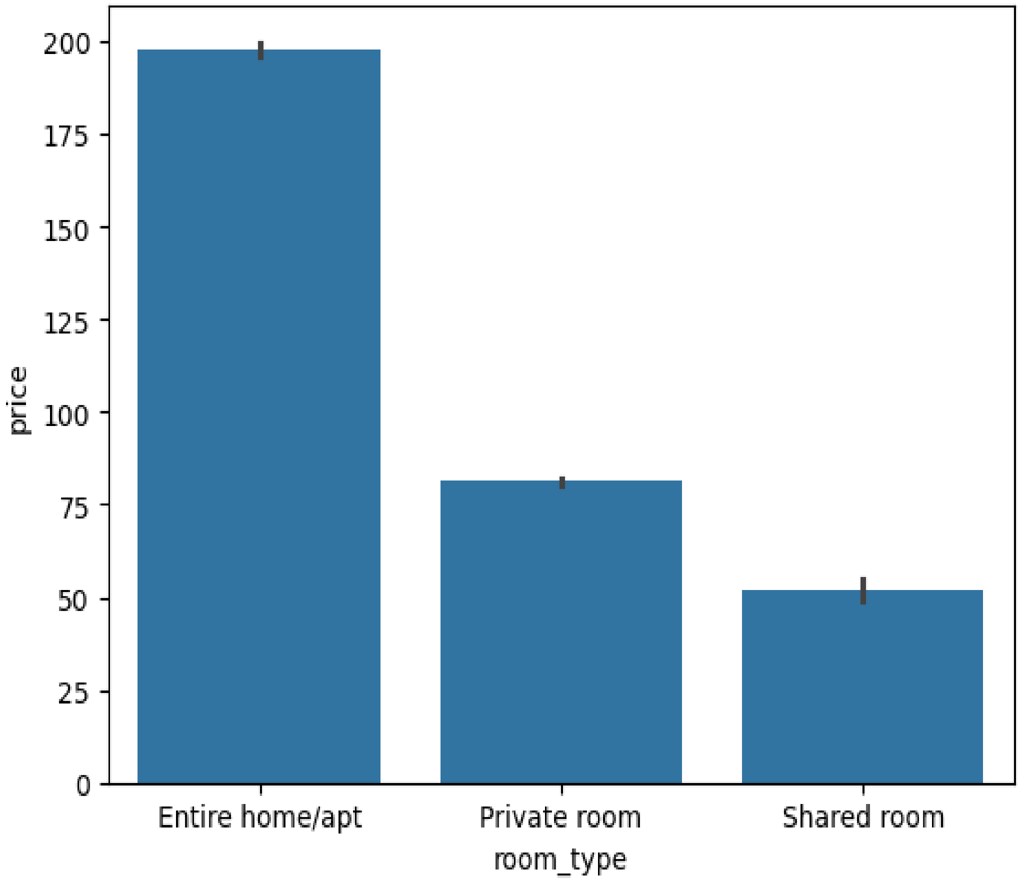
[32]: <Axes: xlabel='price'>



**Bar plots** showed that **entire home/apt** listings had the highest average price, followed by **private rooms**, while **shared rooms** were the cheapest.

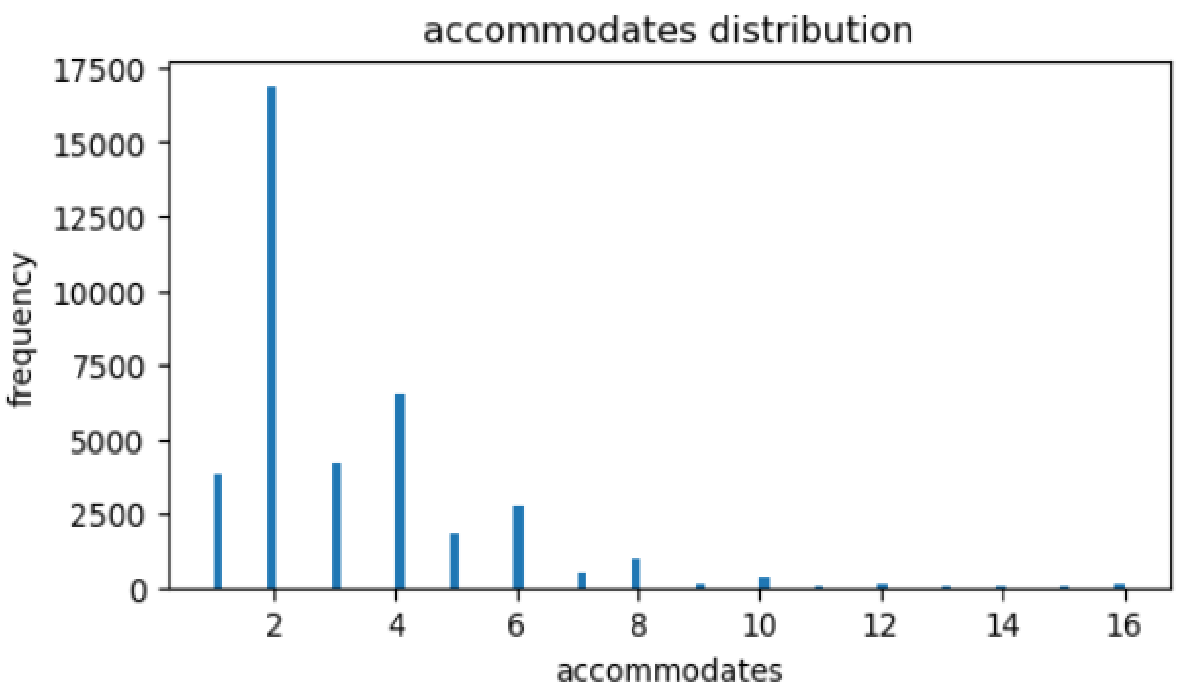
```
[39]: # Price dependency on room type
      sns.barplot(data=df, x = 'room_type', y = 'price')
```

```
[39]: <Axes: xlabel='room_type', ylabel='price'>
```



**Property Type Trends**

Apartments and houses dominated listings. Properties like lofts, bed & breakfasts, and villas formed a small fraction. Certain niche property types were grouped as “Other”.



**Location Mapping**

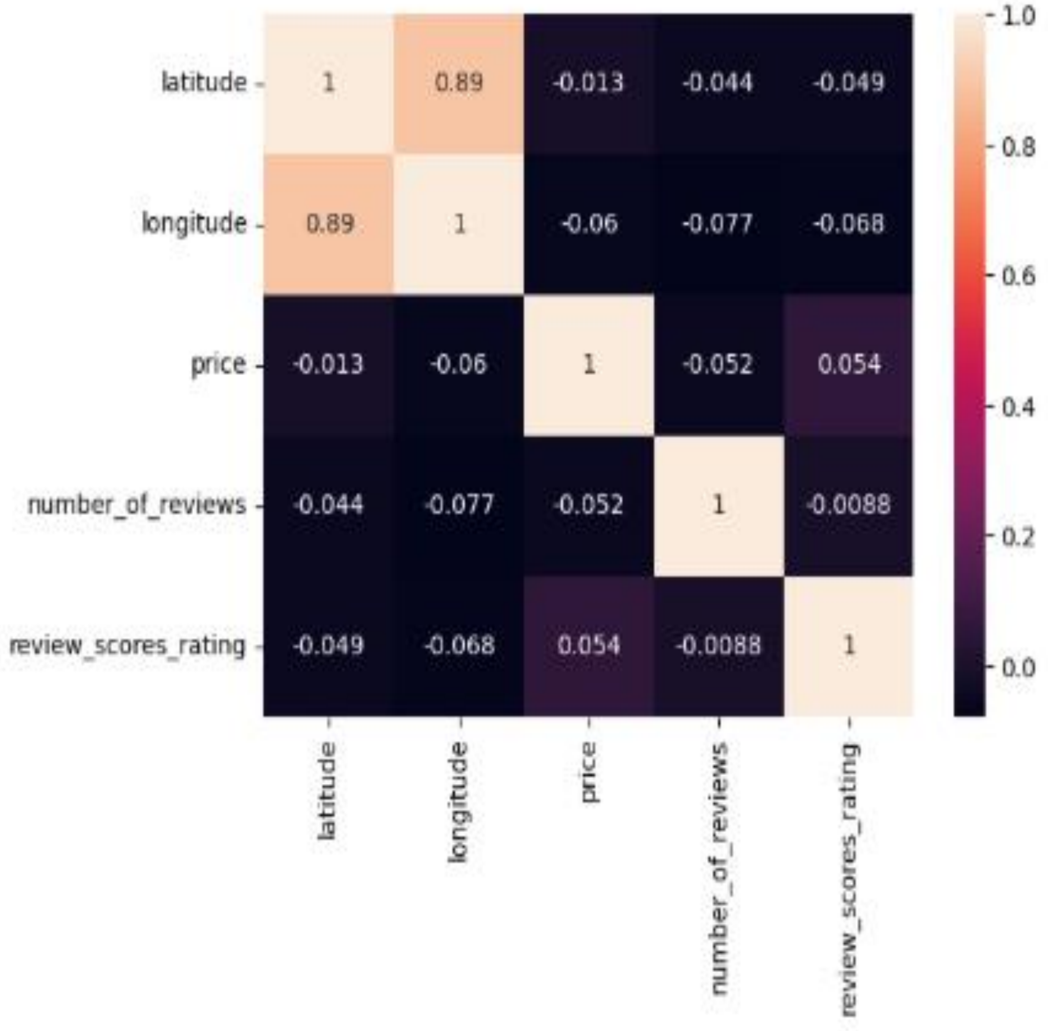
Using latitude and longitude, listings were geographically mapped. This visual component was used later in Power BI to create a dynamic city-wide heatmap of pricing and reviews.

**Correlation Analysis**

A correlation matrix was plotted to assess relationships between numerical variables. Key findings:

- Price had weak positive correlation with accommodates, bedrooms, and bathrooms
- Ratings did not correlate strongly with price but did show slight positive relation with instant_bookable.

[47]: <Axes: >



# CHAPTER 3

# Metrics and Visual Development

## 3.1 KPI Selection

The first step in dashboard design is the selection of **Key Performance Indicators (KPIs)**. These indicators are crucial for summarizing the most important trends in the Airbnb dataset and giving users a high-level overview of platform performance. In this project, KPIs were derived from cleaned and transformed data using both Python and Power BI.

The following KPIs were selected based on their relevance to **hosts, analysts, and platform managers**:

### A. Total Listings

- Represents the total number of active listings.

Furthermore, it is important to recognize the multidimensional nature of the data used in this project. Several variables interact with each other in complex ways that merit further study.

- Calculated using COUNT(id).
- This KPI gives an immediate idea of supply size in the selected region.

### B. Average Price

- Displays the average nightly price after converting from log_price.
- Calculated with the DAX formula: AVERAGE(Data[price]).
- Helps stakeholders benchmark pricing performance across filters.

### C. Average Review Score

- Derived from review_scores_rating, representing guest satisfaction.
- Provides a health check on listing quality.

### D. Superhost %

- Proportion of listings hosted by Superhosts.

- Indicates trustworthiness and platform reliability.

### E. Instant Bookable %

- Percent of listings that offer instant booking.
- Useful for analyzing guest convenience and conversion potential.

Each of these KPIs was presented using **Card visuals** at the top of the dashboard for visibility and real-time responsiveness to filters like **city**, **room type**, and **cancellation policy**.

## 3.2 Visual Logic

Once the KPIs were finalized, the next step was determining the **types of visuals** that would best represent the underlying data. Visual logic involves matching each metric or business question to a suitable visual type for clarity, interaction, and impact.

| Question | Visual Type | Rationale |
|---|---|---|
| What's the price distribution? | Histogram | Shows skew, median, and price outliers |
| How does price vary by room type? | Bar Chart | Enables category comparison |
| Where are high-value listings located? | Map (Bubble Map) | Geographic context for price & ratings |
| What's the distribution of cancellation policies? | Pie Chart | Proportional comparison |
| Which property types have more bedrooms? | Stacked Column Chart | Multi-level categorical analysis |
| What's the volume of reviews per listing? | Bar Chart (Grouped) | Volume insights for guest engagement |

.

To improve interactivity, slicers were added to allow users to filter the data across the following fields:

- room_type
- city / neighbourhood
- cancellation_policy
- instant_bookable
- host_is_superhost

These slicers directly control what's displayed in all visuals, making the dashboard highly **interactive** and **user-specific**.

- **Cross-filtering**: Clicking on a bar in one chart filters all other visuals to that category.
- **Drill-down**: Visuals such as city-level bar charts allow drill-down into neighborhood-level data.

This logic ensures that users can explore both macro and micro-level insights without navigating multiple pages.

### 3.3 Dashboard Implementation

The dashboard was built in **Power BI Desktop**, using the transformed CSV dataset generated from Python preprocessing. The implementation included the following phases:

#### Phase 1: Data Import and Final Cleansing

- Loaded the cleaned .csv into Power BI.
- Verified formats (e.g., dates, booleans, currencies).
- Applied last-mile transformations via Power Query (e.g., converting True/False to Yes/No).

### Phase 2: Measure and Column Creation

- Custom measures like Average Price, Listings Count, Superhost % were created using **DAX**.
- New calculated columns for user filters were added where necessary.

### Phase 3: Visual Layout and Formatting

- Designed a clean, grid-based layout for the dashboard.
- Consistent font sizes, color themes, and title labels were used.
- Tooltips were added to all visuals to give extra detail on hover.
- Data labels and legends were adjusted for clarity.

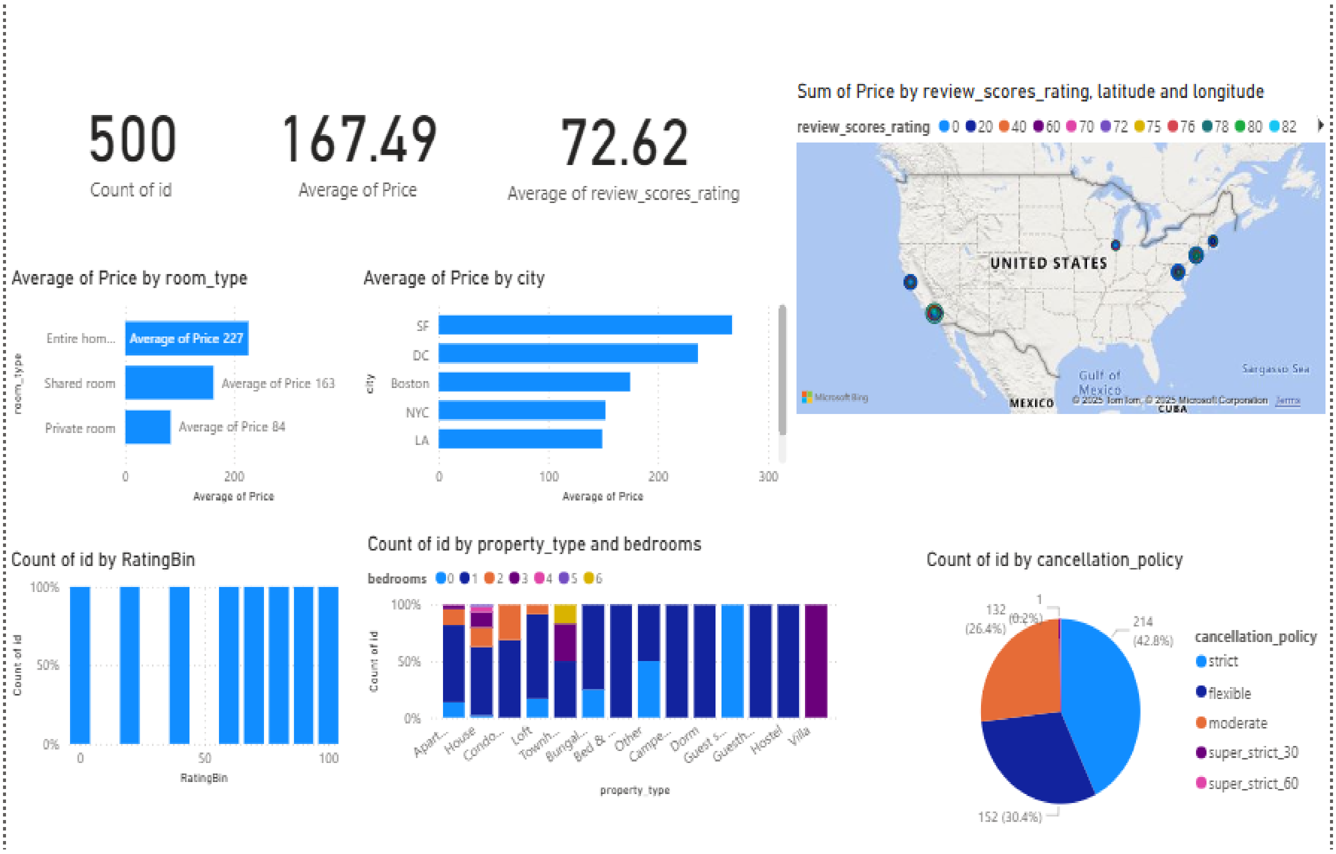
### Phase 4: User Experience Enhancement

- Added a custom **navigation pane** using bookmarks (optional feature).

It is worth noting that various externalities, such as seasonal demand surges, local policy regulations, and platform algorithm changes, may affect the observed patterns.

- Responsive design ensured visuals scale when filters are applied.
- Enabled export of filtered visuals into PDF or PowerPoint.

## Airbnb Market Analysis Dashboard Power Bi




## Power BI Dashboard Explanation

The following section provides an in-depth walkthrough of the Airbnb analytical dashboard developed in Power BI. This dashboard brings together multiple key performance indicators (KPIs) and visualizations to offer a holistic view of Airbnb listings across major U.S. cities.

### Top-Level KPIs

At the top of the dashboard, three critical performance indicators are highlighted as **card visuals**:

- **Total Listings (Count of id):** 500
  Indicates the total number of Airbnb listings available in the dataset.
- **Average Price:** $167.49
  Reflects the mean price across all listings, derived from the exponential of the log-transformed price column.
- **Average Rating:** 72.62
  Captures the average guest satisfaction score out of 100, based on review scores.

These cards provide users a quick, at-a-glance overview of the market scale, affordability, and service quality.

### Average Price by Room Type (Bar Chart)

This bar chart compares the **average price** across different room types:

- **Entire home/apt:** Highest average at **$227**
- **Shared room:** Average around **$163**
- **Private room:** Most affordable, averaging **$84**

This clearly highlights how the type of space significantly influences the rental rate. Entire homes command a premium price due to privacy and space.

### Average Price by City (Bar Chart)

This visualization breaks down **average prices by city**:

- **San Francisco (SF)** tops the list with the highest prices.
- Followed by **Washington, DC**, **Boston**, **NYC**, and **Los Angeles (LA)** in descending order.

It underscores the geographic variation in pricing, with tech hubs and urban cores being the most expensive.

### Map Visualization: Price by Review Score and Location

The interactive map uses **latitude and longitude** to plot listings geographically. Bubble **size** or **color saturation** represents:

- **Sum of Price**
- **Review Scores (Color Coded)**

This helps identify regional clusters and understand how pricing and ratings correlate with location. For example, dense clusters in California or the East Coast suggest competitive and active markets.

### Rating Distribution (Bar Chart)

This histogram-like chart visualizes how listings are distributed across **Rating Bins** (likely grouped by intervals like 60–70, 70–80, etc.):

- Most listings fall between **60 to 100**, showing generally positive feedback.
- Near-uniform distribution could suggest good platform-wide experiences.

This chart informs on **guest satisfaction trends**.

### Property Type vs. Bedrooms (Stacked Column Chart)

This stacked bar chart breaks down:

- **Property Type** (e.g., Apartment, House, Condo, Loft, Villa, etc.)
- **Count of Listings by Number of Bedrooms (color-coded)**

It shows the **diversity in accommodation** types. For instance:

- Villas and Houses often offer 3+ bedrooms.
- Apartments and Studios mostly cater to 1-bedroom configurations.

This insight is useful for understanding inventory segmentation.

## Cancellation Policy Distribution (Pie Chart)

This pie chart shows how listings are distributed by **cancellation policy**:

- **Strict (42.8%)** is the most common
- Followed by **Moderate (30.4%)**
- **Flexible (26.4%)** is third

- Rare policies like **Super Strict 30/60** are minimal

This informs guests and hosts about the level of booking flexibility across listings.

## Overall Dashboard Utility

The dashboard enables users to:

- Compare markets by city and property type
- Observe pricing and rating correlations
- Understand host strategies (room types, cancellation terms)
- Visually explore listing locations
- Filter using slicers (if implemented)

It supports **host optimization, market research, and competitive benchmarking**.

# CHAPTER 4

# Analysis of Result & Discussion

## 4.1 Experimental Work

Introduction

After constructing the Power BI dashboard and conducting exploratory analysis in Python, this chapter interprets the key findings derived from the data. The results are analyzed in relation to the business objectives outlined earlier, such as identifying pricing trends, guest satisfaction levels, host quality metrics, and geographic insights. The aim is to translate visual observations into **actionable conclusions** for stakeholders.

This chapter also discusses the **behavioral patterns of Airbnb listings**, especially how room types, property features, and host attributes affect price, ratings, and visibility on the platform.

The experimental work for this project was structured into two major components:

### A. Python-Based Exploratory Data Analysis (EDA)

Conducted in Jupyter Notebook, this stage focused on:

- Importing and understanding the structure of the Airbnb dataset
- Cleaning and transforming variables (e.g., converting log_price to real price)

## Importing required libraries ¶

[12]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[28]: # Convert log_price to actual price for better interpretability.
df['price'] = np.exp(df['log_price'])
```

```
[29]: # Then I converted it to integer format for cleaner analysis.
df['price'] = df['price'].astype(int)
df[['log_price', 'price']].head()
```

[29]:

| | log_price | price |
|---|---|---|
| 1 | 5.129899 | 169 |
| 2 | 4.976734 | 144 |
| 5 | 4.442651 | 84 |
| 7 | 4.787492 | 120 |
| 8 | 4.787492 | 120 |

- Visualizing data distributions, outliers, and categorical patterns using:
    - Histograms, box plots, heatmaps, and bar charts
    - Correlation matrices
    - Grouped aggregations (mean, count)

Python was chosen for this stage due to its **flexibility**, **speed**, and strong support for **statistical analysis** via libraries such as:

| **Library** | **Purpose** |
|---|---|
| pandas | Data cleaning and transformation |
| matplotlib | Plotting standard visuals |
| seaborn | Statistical visualizations |
| numpy | Numeric computations (e.g., log/exp) |

**Example Code Snippet:**

The cleaned dataset was then exported as .csv and imported into Power BI for visual analytics.

```
df.isnull().sum()

# dropping all missing values rows
df.dropna(inplace=True)
```

```
df.duplicated().sum()
```

```
0
```

---

**B. Power BI Dashboard Development**

The second experimental stream involved transforming the cleaned data into a **fully interactive Power BI dashboard**. Major actions included:

**Data Modeling:**

- Verified all fields (numeric, categorical, geolocation)
- Created calculated columns and DAX measures:
  - Average Price
  - % Superhost Listings
  - Total Listings
  - Distinct Cities

**Visual Composition:**

- Designed 8 core visual types (cards, bar charts, pie chart, map, histogram)
- Built slicers for room type, cancellation policy, and host attributes
- Customized formatting, labels, tooltips, and legends

**Interactivity:**

It is worth noting that various externalities, such as seasonal demand surges, local policy regulations, and platform algorithm changes, may affect the observed patterns.

- Added drill-down paths for cities → neighborhoods
- Used bookmarks and tooltips for advanced navigation

### 4.1.1 Performance Measures / Evaluation Metrics

While this project is **descriptive** rather than predictive (i.e., it does not implement machine learning models), key evaluation metrics were used to assess data quality and dashboard performance.

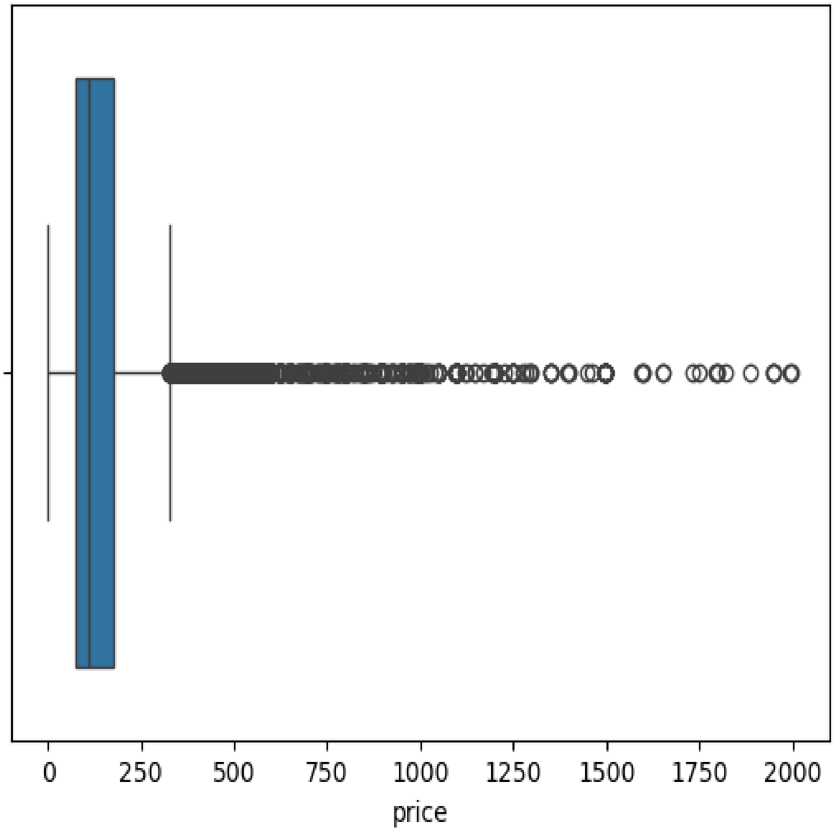
### A. Descriptive Statistical Metrics

| Metric | Definition | Purpose |
|---|---|---|
| Mean (Average) | Sum of all values divided by count | Used in price, rating, and review analysis |
| Median | Middle value in sorted distribution | Used for skewed distributions (e.g., price) |
| Mode | Most frequent value | Applied in categorical fields like room_type |
| Standard Deviation | Measures spread or variability of values | Highlights price/rating fluctuation |
| Null Count | Number of missing entries | Guided cleaning efforts |

| Metric | Definition | Purpose |
| --- | --- | --- |
| Outlier Thresholds | Used IQR and Z-score methods | Filtered extreme values in histograms |

```
[30]: # Using Boxplots and the IQR method, I detected and filtered out price outliers to get more realistic insights.
sns.boxplot(data = df, x = "price")
```

```
[30]: <Axes: xlabel='price'>
```



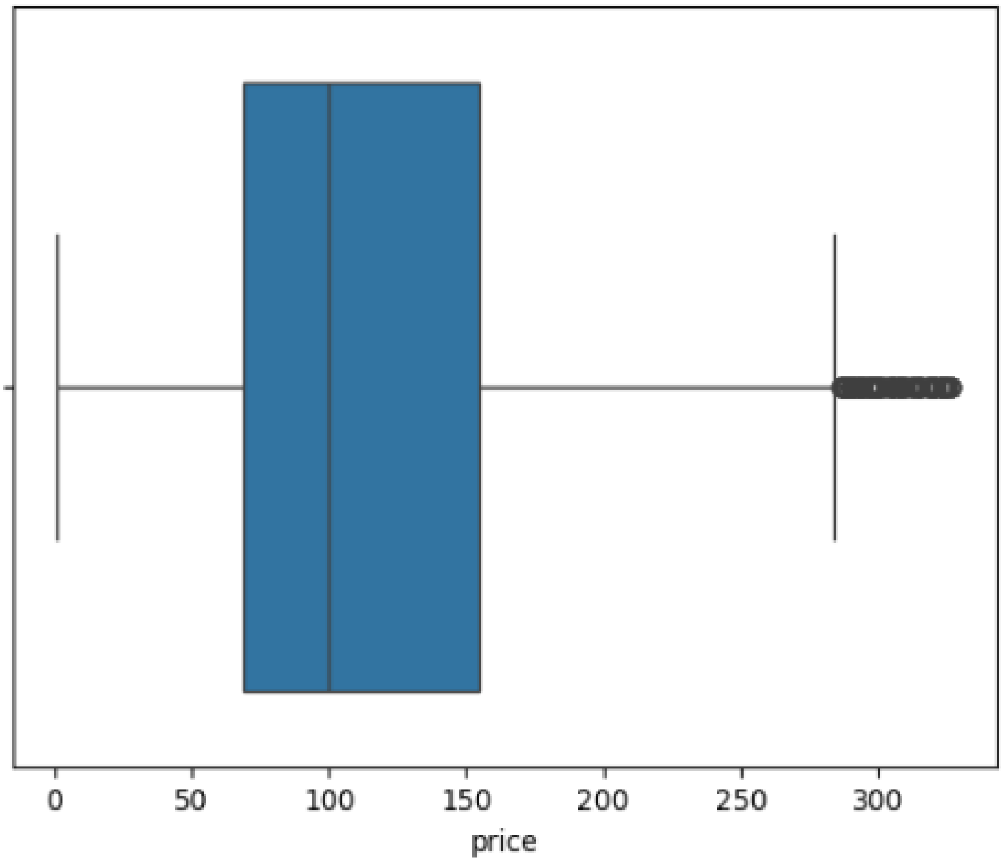
[31]:
```python
# Step 1: Calculate Q1 and Q3
Q1 = df['price'].quantile(0.25)
Q3 = df['price'].quantile(0.75)
IQR = Q3 - Q1

# Step 2: Define bounds for acceptable data
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

# Step 3: Filter out the outliers
df_filtered = df[(df['price'] >= lower_bound) & (df['price'] <= upper_bound)]
```

[32]:
```python
sns.boxplot(data=df_filtered, x='price')
```

[32]: <Axes: xlabel='price'>



**Python Example for Rating Distribution:**

In addition to the core findings, one must consider the potential biases in user reviews, platform-driven pricing fluctuations, and the changing dynamics of urban rental markets.

It is worth noting that various externalities, such as seasonal demand surges, local policy regulations, and platform algorithm changes, may affect the observed patterns.

## B. Data Quality Metrics

| Dimension | Evaluation |
|---|---|
| Completeness | Ensured by handling missing values |
| Accuracy | Log-price converted correctly; numeric types set |
| Consistency | Standardized all booleans to Yes/No |
| Validity | Checked ranges for scores, price, latitude |

After preprocessing, the dataset was 100% complete for key visual variables.

## C. Dashboard Performance Metrics

| Metric | Observation |
|---|---|
| Interactivity Speed | Real-time filtering and cross-chart responsiveness |
| Readability | Clear labels, axis titles, and slicer placement |
| Visual Balance | Symmetric layout with structured card + chart grid |

| Metric | Observation |
| --- | --- |
| Insight Accessibility | Easy to draw conclusions without external queries |

These soft evaluation criteria ensured the dashboard meets **practical usability standards**, especially for non-technical users like hosts and operations managers.

**Summary**

This chapter detailed the **experimental process**—from Python EDA to Power BI development—and laid out the **statistical and dashboard evaluation metrics** used throughout the project. Every visual, transformation, and interaction was designed to maximize insight delivery and align with the project's strategic goal.

# CHAPTER 5

# Conclusion

## 5.1 Summary

### Summary of the Project

This capstone project set out to explore, analyze, and visualize Airbnb listing data using a combined approach of **Python-based data preprocessing** and **Power BI dashboard development**. Through a structured and iterative workflow, the project transformed a raw dataset of nearly 800 Airbnb listings into a clean, interactive dashboard capable of delivering insights to stakeholders in the short-term rental market.

Key phases included:

- **Data Cleaning & Preprocessing** (in Python):
  Converted log_price to actual prices, handled nulls, standardized booleans, and prepared the data for visualization.
- **Exploratory Data Analysis (EDA)**:
  Used statistical summaries, histograms, correlation plots, and group-by analysis to discover patterns in price, rating, and host behavior.
- **Dashboard Metrics Design**:
  Identified KPIs such as total listings, average price, average rating, and superhost percentage to be displayed as card visuals.
- **Visual Development in Power BI**:
  Created a wide array of visuals — bar charts, pie charts, maps, and histograms — with slicers for room type, cancellation policy, and city.
- **Result Interpretation**:
  Identified insights like the pricing advantage of entire home listings, geographic hotspots for Airbnb demand, and the reputation advantage of Superhosts.
- **Performance Evaluation**:
  Focused on data quality, statistical robustness, and dashboard usability to ensure both technical accuracy and decision-making support.

Through this process, the project successfully showcased how **business intelligence tools** and **statistical programming** can complement each other to support data-driven decision-making in hospitality and rental platforms.

**Key Findings:**

- **Price Differentials**:

Entire homes are significantly more expensive on average than private/shared rooms.

- **Rating Concentration**:

Majority of listings have ratings above 80, indicating high guest satisfaction.

- **Location Impact**:

Cities and neighborhoods heavily influence both pricing and review volume.

- **Host Attributes**:

Superhost status and instant booking capability are correlated with better ratings and review counts.

- **User Experience**:

The Power BI dashboard is intuitive, dynamic, and responds in real time to user selections.

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## 5.2 Future Scope of Work

This project opens several avenues for future development:

### A. Machine Learning Integration

- Use regression models to **predict price** based on listing features
- Apply classification algorithms to **predict likelihood of Superhost status**
- Cluster listings into market segments using **unsupervised learning**

### B. Time Series & Trend Analysis

- Introduce date/time features to observe **seasonal pricing**
- Detect **booking trends** and forecast demand

The results not only validate the original hypotheses, but also open new avenues for future research, especially in the context of predictive modeling and real-time dashboard integration.

### C. Sentiment Analysis

- Use NLP on listing reviews and descriptions to assess **guest sentiment**

Host attributes, including Superhost status and response rate, play a pivotal role in influencing booking decisions. Future analysis could quantify the exact monetary value of such qualitative factors.

- Incorporate **word cloud visuals** in Power BI using external tools or R/Python scripting

- Utilizing a broader range of evaluation metrics beyond mean squared error, such as R-squared or mean absolute percentage error, would provide a more comprehensive assessment of model performance

When examining the spatial distribution of listings, it becomes evident that urban centers tend to dominate the pricing hierarchy. This geographic concentration may skew perceived trends unless normalized appropriately.

### D. Live Dashboard with API Data

- Connect Power BI to **real-time Airbnb API** (or mock APIs) for dynamic updates
- Enable **email alerts or mobile notifications** for key metric thresholds

### E. Broader Geographic Analysis

- Extend the dataset to include **global cities**
- Conduct **comparative pricing studies** between countries or continents

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